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Multi-center intervention study on glycohemoglobin (HbA1c) and serum, high-sensitivity CRP (hs-CRP) after local anti-infectious periodontal treatment in type 2 diabetic patients with periodontal disease

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ABSTRACT

The purpose of this study was to examine whether periodontal treatment incorporating topical antibiotic therapy affects on levels of glycohemoglobin (HbA1c) and serum high-sensitivity C-reactive protein (hs-CRP) in type 2 diabetic patients with periodontal disease, and to explore the relationship between CRP and glycemic control. The whole intervention group ($n = 32$), which underwent anti-infectious periodontal treatment, showed only transient reduction in HbA1c levels without any change in hs-CRP, while the control group ($n = 17$) did not show any changes in HbA1c or hs-CRP. Multiple regression analysis of all subjects revealed that BMI and change in hs-CRP correlated significantly with the reduction of HbA1c at 6 months after the periodontal treatment. Based on the results of multiple regression analysis, the intervention group was subdivided into two groups: those in which hs-CRP levels decreased (CRP-D group), and those in which hs-CRP levels unchanged or increased (CRP-N group) ($n = 16$, respectively), and re-analysis was conducted based upon these subgroups. In the CRP-D subgroup, HbA1c was significantly reduced at the end of the study, but it did not decrease in the CRP-N subgroup. The decrease of HbA1c in the CRP-D subgroup following periodontal treatment was significantly greater than that in the CRP-N

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subgroup. BMI of each group remained unchanged in this study at the end of the study. Thus, the results suggested that periodontal treatment with topical antibiotics improves HbA1c through reduction of CRP, which may relate to amelioration of insulin resistance, in type 2 diabetic patients with periodontal disease.

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1. Introduction

Periodontal disease is a chronic infectious disease, triggered by the bacterial biofilm of dental plaque, which results in an inflammatory loss of bone, and both soft and hard tooth-supporting structures [1,2]. It is very often associated in diabetic patients following microangiopathies and macroangiopathies [3]. The prevalence of periodontal disease among individuals with inadequately controlled type 2 diabetes is generally higher than that of non-diabetic individuals [4]. Thus, it is recognized that diabetic patients with poor glycemic control are at increased risk for the incidence and progression of periodontal disease. However, few reports indicated that glycemic control in diabetic patients improved periodontal disease.

On the contrary, there are a large number of reports that have addressed the effects of periodontal disease on glycemic control. A 2-year longitudinal study demonstrated a six-fold increased risk of worsening glycemic control in type 2 diabetic patients with severe periodontitis compared with those with no periodontitis [5]. Intervention studies focusing on the effects of periodontal treatment on glycemic control in diabetic patients with periodontal disease have shown that periodontal treatment with or without adjunctive antibiotic therapy improved metabolic control of diabetes [6–9]. However, other studies found no improvement in glycemic control after periodontal treatment [10]. The reason for the discrepancy regarding efficacy remains unclear at present.

Although type 2 diabetes is a multifactorial metabolic disease, recent evidence suggests that chronic subclinical inflammation plays an important role in the development of type 2 diabetes [11,12]. Elevated circulating levels of subclinical inflammatory markers, such as C-reactive protein (CRP) and interleukin-6 (IL-6) are reported to be significant risk indicators not only for the development of cardiovascular disease, but also for the development or progression of type 2 diabetes [13]. Several reports have demonstrated a significant elevation of serum CRP levels in patients with periodontitis compared with periodontally healthy controls [14–18]. Some intervention studies have suggested that successful anti-infective periodontal treatment can reduce CRP with TNF-α and IL-6 levels in systemically healthy subjects [19–22]. Thus, the possibility exists that reduction of CRP may contribute to the improvement of glycemic control in diabetic patients after periodontal treatment. However, to the best of our knowledge, there have been no studies that investigated the effects of periodontal treatment on the levels of CRP and HbA1c simultaneously in type 2 diabetic patients with periodontal disease.

The purpose of this study was to examine whether anti-infectious periodontal treatment with topical antibiotic therapy affects levels of HbA1c and serum CRP in type 2 diabetic patients with periodontal disease without changes of the treatment of diabetes (drug, diet and exercise) for glycemic control, and to explore the relationship between the CRP and HbA1c.

2. Subjects and methods

2.1. Subjects

A total of 63 type 2 diabetic patients were recruited from diabetic clinics at 5 institutes: Tokyo Medical and Dental University Hospital, Kagoshima University Medical and Dental Hospital, Aichi Gakuin University Dental Hospital, Tokyo Medical University Hospital and Kyoto Prefecture Medical University Hospital. At each institute, patients were selected based on the medical condition inclusion criteria: age of 39–75 years, HbA1c 6.5–10.0%, without severe diabetic complications, no evidence of systemic diseases other than diabetes as a risk factor for periodontitis, no systemic antibiotics during the preceding 3 months, no pregnancy or lactation, no allergy to tetracycline, no smoking, and no modifications in the treatment of diabetes during the preceding 2 months. Then, the patients were transferred to the dental clinic in each institute. Patients for the study were selected based on the dental condition inclusion criteria: at least 11 remaining teeth, at least two pocket sites with probing depth 4 mm or more (indicated as mild to severe periodontitis), no periodontal treatment during the preceding 6 months. Finally, 49 participants fulfilled both medical and dental inclusion criteria.

2.2. Study design

Forty-nine subjects compatible with the medical and dental criteria were randomly allocated to either the periodontal treatment (intervention) or non-periodontal treatment (control) groups by the envelope method. The distribution of subjects to the groups was assigned as 2 in the intervention group and 1 in the control group.

In the intervention group, mechanical debridement of the sub-gingival plaque and calculus was performed using piezoelectric ultra sonic scalers, and 10 mg of minocycline ointment (Periofil®, Showa Yakuhin Co., Tokyo, Japan) was administered topically in every periodontal pocket at the end of each visit. The intensive periodontal treatment was completed over the course of four visits within 2 months. Control group participants received instructions for brushing their teeth, including the use of inter-proximal cleaning aids, such as floss and inter-dental brushes, depending on their individual needs. After the completion of 2 months of intensive period-

ontal treatment, all participants visited the respective medical and dental clinics at 1, 3 and 6 months. In the intervention group, additional periodontal treatment including instructions for brushing, supra- and sub-gingival debridement without topical administration of antibiotics were performed, if necessary. The doses and kinds of anti-diabetic drugs, including oral hypoglycemic drugs and insulin injections and methods of diet and exercise were not changed to assess the real effects of periodontal treatment on blood glucose control. The study protocol was approved by the Ethics Committee for each university hospital, and written informed consent was obtained from each of the participants.

2.3. *Periodontal examination*

All participants underwent a full-mouth clinical periodontal examination at the dental clinic. The periodontal parameters were recorded at baseline, 1, 3 and 6 months following the intensive phase of periodontal treatment. Full-mouth clinical measurements of the probing pocket depth (PPD) and bleeding on probing (BOP) were recorded using a manual probe (PCP-UNC 15, Hu-Friedy Manufacturing Co., Inc., Chicago, USA) at all six sites of each tooth.

2.4. *Detection of serum peripheral parameters*

At the medical clinic, venous blood samples were taken from each patient and analyzed for fasting plasma glucose (FPG), glycohemoglobin (HbA1c) and high-sensitivity C-reactive protein (hs-CRP). FPG was determined by a hexokinase assay. HbA1c was determined by high-performance liquid chromatography (Kyotokagaku Co., Ltd., Kyoto, Japan). Serum was immediately stored at −70 °C. Serum hs-CRP was measured using a latex particle-enhanced immunonephelometric method. FPG, HbA1c and hs-CRP measurements were performed at baseline, and 1, 3 and 6 months following the intensive phase of periodontal treatment.

2.5. *Statistical analysis*

Statistical analysis was performed using the JMP® (SAS Institute Inc., Cary, NC, USA) software program. Changes of all parameters from baseline to 1, 3 and 6 months were compared using Wilcoxon signed-rank test. Mann–Whitney *U*-test was used to compare the changes of parameters between the intervention and control groups, and the CRP-decreased (CRP-D) and CRP-unchanged or increased (CRP-N) groups of subanalysis for the intervention group. Multiple regression analysis was performed to examine the effect of age, gender, baseline body mass index (BMI), duration of diabetes mellitus, baseline HbA1c, baseline hs-CRP, number of teeth, baseline PPD, baseline BOP, change in hs-CRP at 1 month (Delta hs-CRP), change in PPD at 1 month (Delta PPD), change in BOP at 1 month (Delta BOP) and intervention of periodontal treatment on the change in HbA1c at 6 months.

3. Results

3.1. *Patient profile*

The analyzed participants included 32 patients in the intervention group and 17 patients in the control group. Table 1A presents the demographics of the participants at

Table 1 – Demographics of the participants at baseline.

	A. All participants		B. Intervention group	
	Intervention[a] (*n* = 32)	Control[a] (*n* = 17)	CRP-D[a] (*n* = 16)	CRP-N[a] (*n* = 16)
Age (years)				
Mean ± S.D. (median)	60.3 ± 9.9 (60)	59.0 ± 4.8 (59)	59.3 ± 10.6 (58)	61.3 ± 9.0 (63)
Gender (*n*)				
Male/female	21/11	6/11	13/3	8/8
BMI				
Mean ± S.D. (median)	24.1 ± 4.1 (23.5)	25.8 ± 4.8 (24)	24.3 ± 4.9 (22.8)	23.8 ± 3.0 (23.6)
Duration (years)				
Mean ± S.D. (median) [range]	11.3 ± 6.4 (6) [3–25]	8.8 ± 7.5 (9.0) [1–30]	8.8 ± 4.9 (6) [3–22]	12.4 ± 7.2 (6) [5–25]
Diabetic treatment				
Diet only/oral hypoglycemic/insulin	1/15/16	2/12/3	0/7/9	1/8/7
Number of teeth (*n*)				
Mean ± S.D. (median)	23.6 ± 5.0 (25)	24.1 ± 2.8 (24)	22.9 ± 5.7 (24)	24.2 ± 4.0 (25)
HbA1c (%)				
Mean ± S.D. (median)	7.2 ± 0.9 (7.1)	6.9 ± 0.9 (6.7)	7.1 ± 0.9 (7.0)	7.4 ± 0.8 (7.2)
FPG				
Mean ± S.D. (median)	152 ± 53 (149)	133 ± 35 (128)	146 ± 56 (149)	159 ± 53 (149)
hs-CRP (ng/ml)				
Mean ± S.D. (median)	1858 ± 2531 (725)	2278 ± 2593 (2025)	2523 ± 3141 (1150)	1192 ± 1569 (400)

There are no significant differences between intervention and control groups, and between CRP-D and CRP-N groups in all parameters.
[a] Groups.

Table 2 – Periodontal status at baseline and follow-up period.

Group	Baseline	1 month	3 months	6 months
A. All participants				
Control group (n = 17)				
PPD (mm)	2.8 ± 0.9	2.6 ± 0.6	2.6 ± 0.7	2.6 ± 0.7*
BOP (%)	24.9 ± 14.7	23.3 ± 16.3	20.1 ± 17.1	18.8 ± 18.1*
Intervention group (n = 32)				
PPD (mm)	3.0 ± 0.9	2.3 ± 0.6§,‡	2.2 ± 0.5§,‡	2.2 ± 0.5§,‡
BOP (%)	36.5 ± 22.5	11.5 ± 10.8§,‡	13.4 ± 12.5§,‡	12.0 ± 11.4§,‡
B. Intervention group				
CRP-D group (n = 16)				
PPD (mm)	3.3 ± 1.1	2.5 ± 0.7§	2.3 ± 0.5§	2.5 ± 0.6§
BOP (%)	46.6 ± 23.7†	15.5 ± 12.3§,†	17.4 ± 14.4§	14.5 ± 13.5§
CRP-N group (n = 16)				
PPD (mm)	2.7 ± 0.6	2.1 ± 0.4§	2.1 ± 0.4§	2.0 ± 0.3§
BOP (%)	26.4 ± 15.8	7.5 ± 6.9§	8.8 ± 7.6§	9.5 ± 8.3§

Values are given as mean ± S.D. PPD: probing pocket depth. BOP: bleeding on probing.
* Statistically significant decrease compared with baseline ($p < 0.05$).
§ Statistically significant decrease compared with baseline ($p < 0.01$).
‡ Statistically significant decrease compared with control group ($p < 0.05$).
† Statistically significant decrease compared with CRP-N group ($p < 0.05$).

baseline. Age, gender, BMI, duration of diabetes mellitus, method of diabetic treatment, number of teeth, HbA1c, FPG and hs-CRP values showed no significant differences between the intervention and control groups.

As shown in Table 2A, which presents the periodontal parameters of the participants, there were no significant differences in the periodontal status, including PPD and BOP, between the intervention and control groups at baseline.

3.2. *Effect of periodontal treatment on the periodontal status*

In the whole intervention group, both PPD and BOP improved significantly at 1 month following the intensive periodontal treatment compared with those at baseline, and these improvements were maintained throughout the study period (Table 2A). In the control group, PPD and BOP were slightly reduced at 6 months compared with those at baseline (Table 2A). PPD and BOP were significantly lower in the intervention group compared with the control group at 1 month after the intensive periodontal treatment and these significant differences continued until the end of the study, which indicated that periodontal treatment in the intervention group was effective (Table 2A).

3.3. *Metabolic assessments*

The levels of HbA1c in the whole intervention and control groups are shown in Fig. 1a. The HbA1c levels in the intervention and control groups were not significantly different at baseline (7.2 ± 0.9% in the intervention group vs. 6.9 ± 0.9% in the control group). One month after the intensive periodontal treatment, a significant reduction in the HbA1c was observed in the intervention group (Fig. 1a). The reduced HbA1c levels in the intervention group tended to continued thereafter. However, the HbA1c levels at 3 or 6 months were not statistically significant compared with the baseline HbA1c level (Fig. 1a). In the control group, HbA1c levels did not show any significant changes in HbA1c levels during the follow-up period.

Serum hs-CRP levels in the whole intervention and control groups are shown in Fig. 1b. There were no significant differences in serum hs-CRP levels between the intervention and control groups at baseline 1, 3 and 6 months. No significant changes were observed in both groups during the follow-up period. There were no significant differences between the intervention and control group at each point during the follow-up period.

BMI and FPG remained unchanged significantly during the whole study in the control and intervention groups.

3.4. *Multiple regression analysis*

To further analyze the effect of periodontal treatment on glycemic control, multiple regression analysis was applied to examine the correlation of the decrease of HbA1c levels from the baseline to the 6-months, with the 13 variables of age, gender, baseline BMI, duration of diabetes mellitus, baseline HbA1c levels, baseline hs-CRP levels, number of teeth, baseline PPD, baseline BOP, change in hs-CRP between baseline and 1 month after, change in PPD between baseline and 1 month after, change in BOP between baseline and 1 month after and intervention of periodontal treatment. As shown in Table 3, baseline BMI and change in hs-CRP were correlated significantly with the change in HbA1c levels during 6 months in all participants ($p = 0.03$ for BMI, $p = 0.03$ for change in hs-CRP).

3.5. *Effect of periodontal treatment on the periodontal status in the CRP-decreased (CRP-D) and the CRP unchanged or increased (CRP-N) groups*

Based on the results of regression analysis, the intervention group was subdivided into two groups by the change in hs-CRP, and sub-analysis was conducted. Out of 32 patients in the

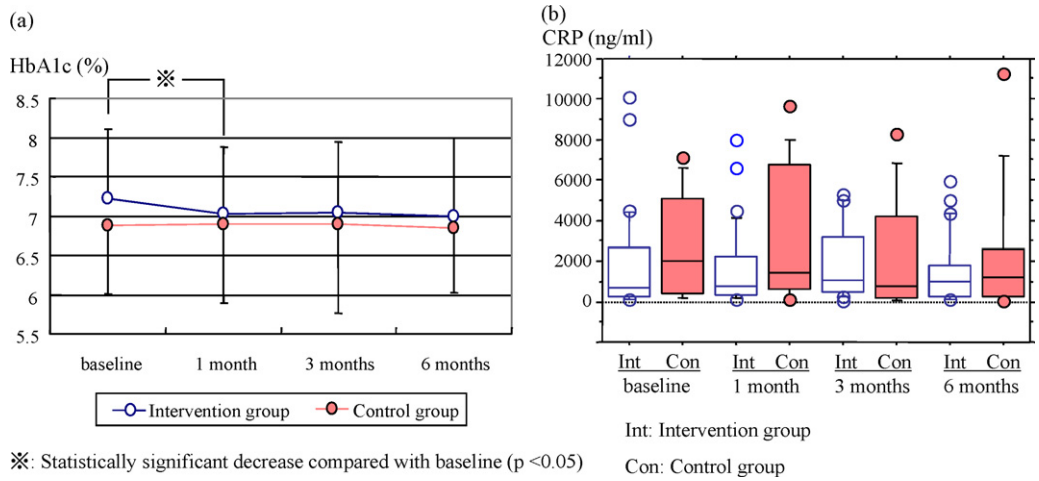

Fig. 1 – The levels of HbA1c and serum CRP at baseline and during the follow-up period in the intervention and control groups. (a) Shows the mean HbA1c levels of the intervention and control groups at baseline and during the follow-up period. There were no significant differences between the intervention and control groups at baseline, 1, 3 and 6 months. No significant differences were observed in HbA1c between the baseline and follow-up period in the control group. There were significant differences in HbA1c between the baseline and 1 month after periodontal treatment in the intervention group. (b) Shows the serum concentration of the high-sensitivity C-reactive protein (hs-CRP) of the intervention and control groups at baseline and during the follow-up period. Box plots show the medians, the 25th and 75th percentiles as boxes, and the 10th and 90th percentiles as whiskers. Values outside this range are shown as open circles. There were no significant differences in hs-CRP concentration between the intervention and control groups. No significant difference was observed in hs-CRP concentration between the baseline and the follow-up period in the intervention or control groups.

intervention group, serum hs-CRP was decreased in 16 patients (CRP-D group), whereas the other patients showed unchanged or increased their serum hs-CRP at 1 month following periodontal treatment (CRP-N group). We re-analyzed the results by comparing the results between the CRP-D and CRP-N groups. As shown in Table 1B, there were no differences in age, gender, baseline BMI, duration of diabetes mellitus, method of diabetic treatment, number of teeth, HbA1c, FPG and hs-CRP at baseline between the CRP-D and CRP-N groups. In addition, PPD showed no differences between the CRP-D and CRP-N groups at baseline. However, BOP was significantly higher in the CRP-D group compared with the CRP-N group at baseline, indicating that active inflammation was more prominent in the CRP-D group than in the CRP-N group (Table 2B). PPD was decreased significantly in both the CRP-D and CRP-N groups at 1 month and continued throughout follow-up period. There were no significant differences in PPD between the CRP-D and CRP-N groups during the follow-up period (Table 2B). BOP significantly decreased in both CRP-D and CRP-N groups at 1 month and it continued during the follow-up period. Although significant differences in BOP were observed at 1 month between the CRP-D and CRP-N groups ($p < 0.05$), but the differences disappeared at 3 and 6 months (Table 2B).

As shown in Fig. 2b, serum hs-CRP in the CRP-D group decreased significantly compared with baseline hs-CRP at 1 and 6 months following periodontal treatment. On the other hand, serum hs-CRP in the CRP-N group remained unchanged during the follow-up period.

Serum HbA1c at baseline was not different between the CRP-D and CRP-N groups. The serum HbA1c in the CRP-D group significantly decreased at 6 months, but it remained unchanged in the CRP-N group during the follow-up period (Fig. 2a). A significant difference between the CRP-D and CRP-N groups was observed at 6 months.

Any significant differences in BMI and FPG were not observed in both subgroups (CRP-D and CRP-N groups) of the intervention group between baseline and the follow-up periods.

Table 3 – Multiple regression model for changes of HbA1c levels between baseline and 6 months.

Standardized partial correlation coefficient	All participants	
	Estimates	p
Age	0.05	0.75
Gender	0.23	0.12
Baseline BMI	0.33	0.03
Duration	−0.27	0.09
Baseline HbA1c	−0.24	0.15
Baseline ln [hs-CRP]	0.28	0.08
Number of teeth	0.03	0.87
Baseline PPD	0.09	0.78
Baseline BOP	−0.04	0.89
Change in hs-CRP	0.33	0.03
Change in PPD	0.40	0.22
Change in BOP	0.004	0.99
Intervention of periodontal treatment	−0.19	0.42

Model $r^2 = 0.65$.

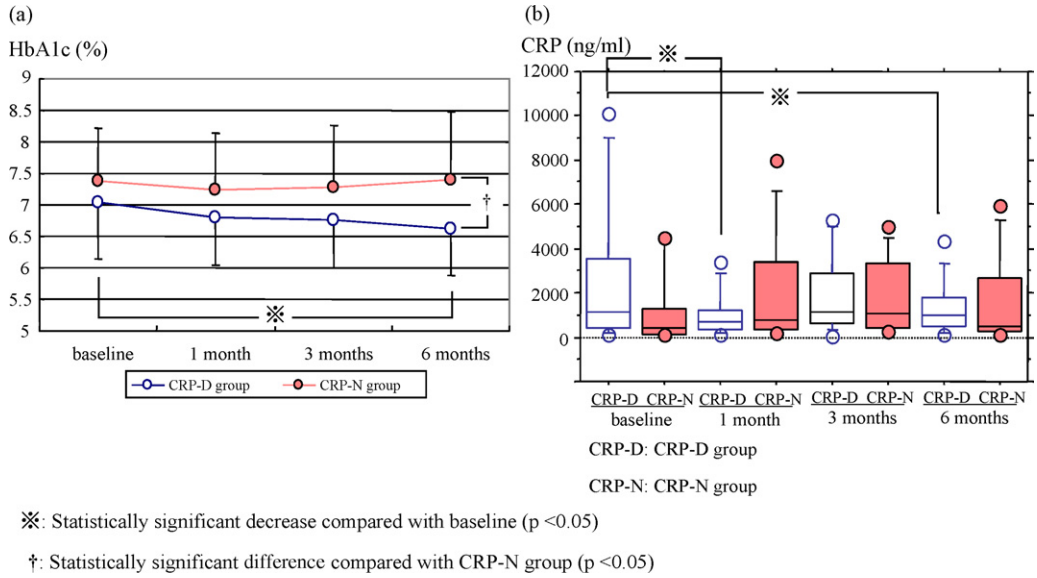

Fig. 2 – The levels of HbA1c and serum CRP at baseline and during the follow-up period in the CRP-decreased (CRP-D) and CRP-unchanged or increased (CRP-N) groups. (a) Shows the HbA1c levels of the CRP-D and CRP-N groups at baseline and during the follow-up period. There were no significant differences in HbA1c between the CRP-D and CRP-N groups at baseline, 1, 3 and 6 months. In the CRP-D group, there were significant differences in HbA1c level between the baseline and 6 months follow-up periods. No significant differences were observed between the baseline and follow-up period in the CRP-N group. (b) Shows the serum concentration of hs-CRP of the CRP-D and CRP-N groups at baseline and during the follow-up period. Box plots show the medians, the 25th and 75th percentiles as boxes, and the 10th and 90th percentiles as whiskers. Outside values are shown as open circles. In the CRP-D group, there were significant differences in hs-CRP between the baseline and the 1- and 6-month follow-up period. No significant changes were observed in concentration of hs-CRP during the whole study in the CRP-N group.

4. Discussion

The main results of the present study are summarized as follows: without any changes in hypoglycemic treatments, including drugs, diet and exercise; (1) periodontal treatment with local antibiotics administration was able to improve glycemic control in type 2 diabetic patients, (2) the improvement of glycemic control was significant when associated with a reduction of hs-CRP, (3) patients with hs-CRP associated with high BOP responded well to periodontal treatment and exhibited reduction of HbA1c, and (4) CRP contributed to the improvement of glycemic control by the treatment of periodontal disease.

Recent evidence has indicated that patients with severe periodontitis exhibit increased serum levels of hs-CRP, as well as proinflammatory cytokines, such as TNF-α and IL-6 [21,23] and these levels decrease after periodontal treatment [21,22]. These results supported that local infection contributes to systemic inflammation [21,24]. In diabetic patients with periodontal disease, Lalla et al. also reported that periodontal treatment with antibiotics reduced hs-CRP [25]. However, it has not been determined whether reduced hs-CRP may relate to reduce HbA1c in diabetic patients since the effects of anti-infectious periodontal treatment on HbA1c, have been inconsistent [6–9,26,27]. Meta-analysis of 10 reliable intervention studies with more than 456 type 2 diabetic patients with periodontal disease demonstrated an average decrease in actual HbA1c, an indicator of glycemic control, of 0.57%, but the reduction was not significant [28].

In this study, a remarkable improvement of periodontal status was observed in all patients after periodontal treatment. However, significant reduction of HbA1c was observed only in diabetic patients in the CRP-D group. Moreover, multiple regression analysis of all subjects revealed that Delta hs-CRP level at 1 month correlated significantly with the reduction of HbA1c at 6 months after periodontal treatment in this study. It suggests that change in hs-CRP level is a contributing factor relating to the changes in HbA1c level. These results suggested that anti-infectious periodontal treatment is effective when associated with a reduction in hs-CRP, and the results may, in part, explain the inconsistent effects of periodontal treatment on glycemic control in diabetic patients.

It is noteworthy that patients in CRP-D group associated with higher bleeding on probing, which indicated active inflammation, responded well to anti-infectious treatment with antibiotics, resulting in reduction of HbA1c in this study. It is consistent with the results that periodontal treatment with antibiotics reduced the levels of hs-CRP [29] and HbA1c [8,30,31] more than that without antibiotics.

What caused the reduction of HbA1c by anti-infectious periodontal treatment in type 2 diabetic patients in this study? Some reports have suggested that improvement of insulin resistance was associated with improvement of glycemic control in type 2 diabetic patients with periodontal disease after periodontal treatment [9,23]. It has been reported that hs-CRP directly induces insulin resistance in type 2 diabetic patients [11,32]. Periodontal diseases also release proinflam-

matory cytokines such as TNF-α and IL-6 [18]. TNF-α and IL-6 have been reported to induce insulin resistance [33,34].

Anti-infectious periodontal treatment ameliorates local inflammation, which reduces production of CRP as well as TNF-α and IL-6, resulting in improvement of insulin resistance [9,23].

In the present study, supportive periodontal treatment including additional periodontal debridement was performed in follow-up period after the intensive periodontal treatment, if necessary. As a result, improved periodontal parameters were maintained during the period of study. This may suggest that appropriate supportive periodontal treatment is needed to maintain improved glycemic control obtained after the intensive periodontal treatment.

The regression analysis in this study revealed that another contributing factor to the effect of anti-infectious periodontal treatment on reduced HbA1c was BMI. However, a mean BMI of each group remained unchanged at the end of this study. Consequently, the result suggested that BMI did not contribute to reduce HbA1c level after the periodontal treatment in this study.

In conclusion, periodontal treatment with topical antibiotics was effective towards the improvement of glycemic control in type 2 diabetic patients with periodontal disease when associated with a decrease in serum hs-CRP. This report indicated the possibility that CRP may play independent contributing role on the reduced HbA1c in diabetic patients by local periodontal treatment with antibiotics.

Conflict of interest

There is no conflict of interest in this study.

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REFERENCES

[1] R.C. Page, S. Offenbacher, H.E. Schroeder, G.J. Seymour, K.S. Kornman, Advances in the pathogenesis of periodontitis: summary of developments, clinical implications and future directions, Periodontology 2000 (14) (1997) 216–248.
[2] B.L. Pihlstrom, B.S. Michalowicz, N.W. Johnson, Periodontal diseases, Lancet 366 (2005) 1809–1820.
[3] B.L. Mealey, T.W. Oates, Diabetes mellitus and periodontal diseases, J. Periodontol. 77 (2006) 1289–1303.
[4] C. Tsai, C. Hayes, G.W. Taylor, Glycemic control of type 2 diabetes and severe periodontal disease in the US adult population, Community Dent. Oral Epidemiol. 30 (2002) 182–192.
[5] G.W. Taylor, B.A. Burt, M.P. Becker, J. Genco, M. Shlossman, W.C. Knowler, et al., Severe periodontitis and risk for poor glycemic control in patients with non-insulin-dependent diabetes mellitus, J. Periodontol. 67 (1996) 1085–1093.
[6] M. Kiran, N. Arpak, E. Unsal, M.F. Erdogan, The effect of improved periodontal health on metabolic control in type 2 diabetes mellitus, J. Clin. Periodontol. 32 (2005) 266–272.
[7] J.E. Stewart, K.A. Wager, A.H. Friedlander, H.H. Zadeh, The effect of periodontal treatment on glycemic control in patients with type 2 diabetes mellitus, J. Clin. Periodontol. 28 (2001) 306–310.
[8] S.G. Grossi, F.B. Skrepcinski, T. DeCaro, D.C. Robertson, A.W. Ho, R.G. Dunford, et al., Treatment of periodontal disease in diabetics reduces glycated hemoglobin, J. Periodontol. 68 (1997) 713–719.
[9] Y. Iwamoto, F. Nishimura, M. Nakagawa, H. Sugimoto, K. Shikata, H. Makino, et al., The effect of antimicrobial periodontal treatment on circulating tumor necrosis factor-alpha and glycated hemoglobin level in patients with type 2 diabetes, J. Periodontol. 72 (2001) 774–778.
[10] M. Christgau, K.D. Palitzsch, G. Schmalz, U. Kreiner, S. Frenzel, Healing response to non-surgical periodontal therapy in patients with diabetes mellitus: clinical, microbiological, and immunologic results, J. Clin. Periodontol. 25 (1998) 112–124.
[11] A. Festa, R. D'Agostino Jr., G. Howard, L. Mykkanen, R.P. Tracy, S.M. Haffner, Chronic subclinical inflammation as part of the insulin resistance syndrome: the Insulin Resistance Atherosclerosis Study (IRAS), Circulation 102 (2000) 42–47.
[12] K. Matsumoto, Y. Sera, Y. Abe, Y. Ueki, T. Tominaga, S. Miyake, Inflammation and insulin resistance are independently related to all-cause of death and cardiovascular events in Japanese patients with type 2 diabetes mellitus, Atherosclerosis 169 (2003) 317–321.
[13] A.D. Pradhan, J.E. Manson, N. Rifai, J.E. Buring, P.M. Ridker, C-reactive protein, interleukin 6, and risk of developing type 2 diabetes mellitus, JAMA 286 (2001) 327–334.
[14] G.D. Slade, S. Offenbacher, J.D. Beck, G. Heiss, J.S. Pankow, Acute-phase inflammatory response to periodontal disease in the US population, J. Dent. Res. 79 (2000) 49–57.
[15] T. Wu, M. Trevisan, R.J. Genco, K.L. Falkner, J.P. Dorn, C.T. Sempos, Examination of the relation between periodontal health status and cardiovascular risk factors: serum total and high density lipoprotein cholesterol, C-reactive protein, and plasma fibrinogen, Am. J. Epidemiol. 151 (2000) 273–282.
[16] B.G. Loos, J. Craandijk, F.J. Hoek, P.M. Wertheim-van Dillen, U. van der Velden, Elevation of systemic markers related to cardiovascular diseases in the peripheral blood of periodontitis patients, J. Periodontol. 71 (2000) 1528–1534.
[17] B. Noack, R.J. Genco, M. Trevisan, S. Grossi, J.J. Zambon, E. De Nardin, Periodontal infections contribute to elevated systemic C-reactive protein level, J. Periodontol. 72 (2001) 1221–1227.
[18] I. Glurich, S. Grossi, B. Albini, A. Ho, R. Shah, M. Zeid, et al., Systemic inflammation in cardiovascular and periodontal disease: comparative study, Clin. Diagn. Lab. Immunol. 9 (2002) 425–432.
[19] J.L. Ebersole, R.L. Machen, M.J. Steffen, D.E. Willmann, Systemic acute-phase reactants, C-reactive protein and haptoglobin, in adult periodontitis, Clin. Exp. Immunol. 107 (1997) 347–352.
[20] K. Mattila, M. Vesanen, V. Valtonen, M. Nieminen, T. Palosuo, V. Rasi, Effect of treating periodontitis on C-reactive protein levels: a pilot study, BMC Infect. Dis. 2 (2002) 30.
[21] F. D'Aiuto, L. Nibali, M. Parkar, J. Suvan, M.S. Tonetti, Short-term effects of intensive periodontal therapy on serum inflammatory markers and cholesterol, J. Dent. Res. 84 (2005) 269–273.

[22] Y. Iwamoto, F. Nishimura, Y. Soga, K. Takeuchi, M. Kurihara, S. Takashiba, et al., Antimicrobial periodontal treatment decreases serum C-reactive protein, tumor necrosis factor-alpha, but not adiponectin levels in patients with chronic periodontitis, J. Periodontol. 74 (2003) 1231–1236.
[23] F. Nishimura, Y. Iwamoto, J. Mineshiba, A. Shimizu, Y. Soga, Y. Murayama, Periodontal disease and diabetes mellitus: the role of tumor necrosis factor-alpha in a 2-way relationship, J. Periodontol. 74 (2003) 97–102.
[24] K. Yamazaki, T. Honda, T. Oda, K. Ueki-Maruyama, T. Nakajima, H. Yoshie, et al., Effect of periodontal treatment on the C-reactive protein and proinflammatory cytokine levels in Japanese periodontitis patients, J. Periodontal Res. 40 (2005) 53–58.
[25] E. Lalla, S. Kaplan, J. Yang, G.A. Roth, P.N. Papapanou, S. Greenberg, Effects of periodontal therapy on serum C-reactive protein, sE-selectin, and tumor necrosis factor-alpha secretion by peripheral blood-derived macrophages in diabetes. A pilot study, J. Periodontal Res. 42 (2007) 274–282.
[26] J.P. Aldridge, V. Lester, T.L. Watts, A. Collins, G. Viberti, R.F. Wilson, Single-blind studies of the effects of improved periodontal health on metabolic control in type 1 diabetes mellitus, J. Clin. Periodontol. 22 (1995) 271–275.
[27] G.T. Smith, C.J. Greenbaum, B.D. Johnson, G.R. Persson, Short-term responses to periodontal therapy in insulin-dependent diabetic patients, J. Periodontol. 67 (1996) 794–802.
[28] S.J. Janket, A. Wightman, A.E. Baird, T.E. Van Dyke, J.A. Jones, Does periodontal treatment improve glycemic control in diabetic patients? A meta-analysis of intervention studies, J. Dent. Res. 84 (2005) 1154–1159.
[29] F. D'Aiuto, M. Parkar, L. Nibali, J. Suvan, J. Lessem, M.S. Tonetti, Periodontal infections cause changes in traditional and novel cardiovascular risk factors: results from a randomized controlled clinical trial, Am. Heart J. 151 (2006) 977–984.
[30] D.C. Rodrigues, M.J. Taba, A.B. Novaes, S.L. Souza, M.F. Grisi, Effect of non-surgical periodontal therapy on glycemic control in patients with type 2 diabetes mellitus, J. Periodontol. 74 (2003) 1361–1367.
[31] P.A.A. O'Connell, M.T. Nomizo, M.C.F. Freitas, F.A. Suaid, S.A. Uyemura, G.L. Trevisan, et al., Effects of periodontal therapy on glycemic control and inflammatory markers, J. Periodontol. 79 (2008) 774–783.
[32] R. Nesto, C-reactive protein, its role in inflammation, Type 2 diabetes and cardiovascular disease, and the effects of insulin-sensitizing treatment with thiazolidinediones, Diabet. Med. 21 (2004) 810–817.
[33] K.T. Uysal, S.M. Wiesbrock, M.W. Marino, G.S. Hotamisligil, Protection from obesity-induced insulin resistance in mice lacking TNF-alpha function, Nature 389 (1997) 610–614.
[34] P.J. Klover, T.A. Zimmers, L.G. Koniaris, R.A. Mooney, Chronic exposure to interleukin-6 causes hepatic insulin resistance in mice, Diabetes 52 (2003) 2784–2789.